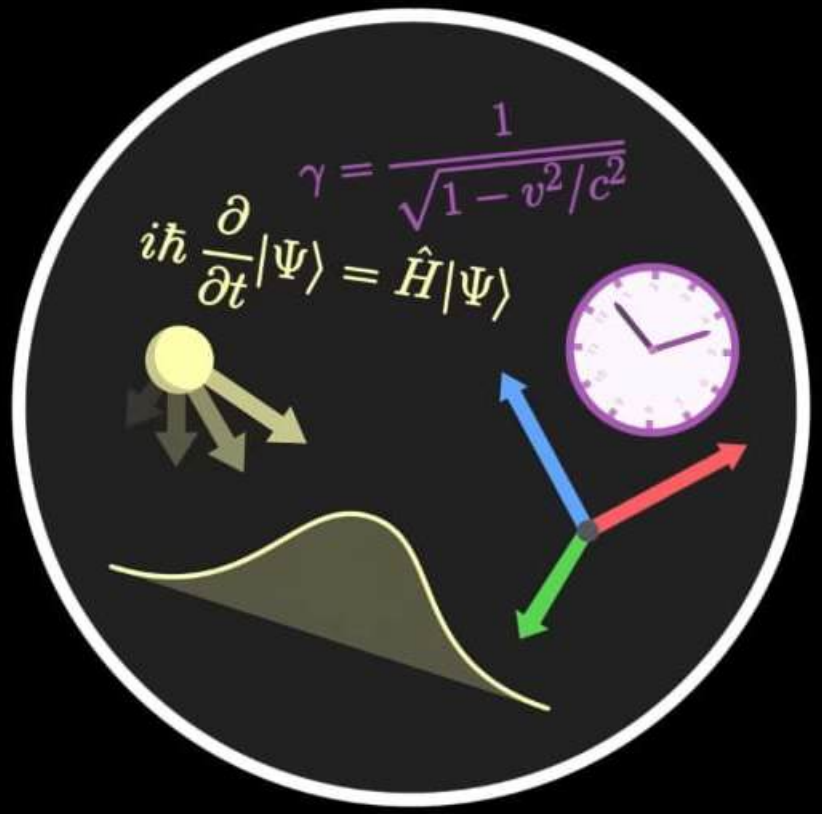
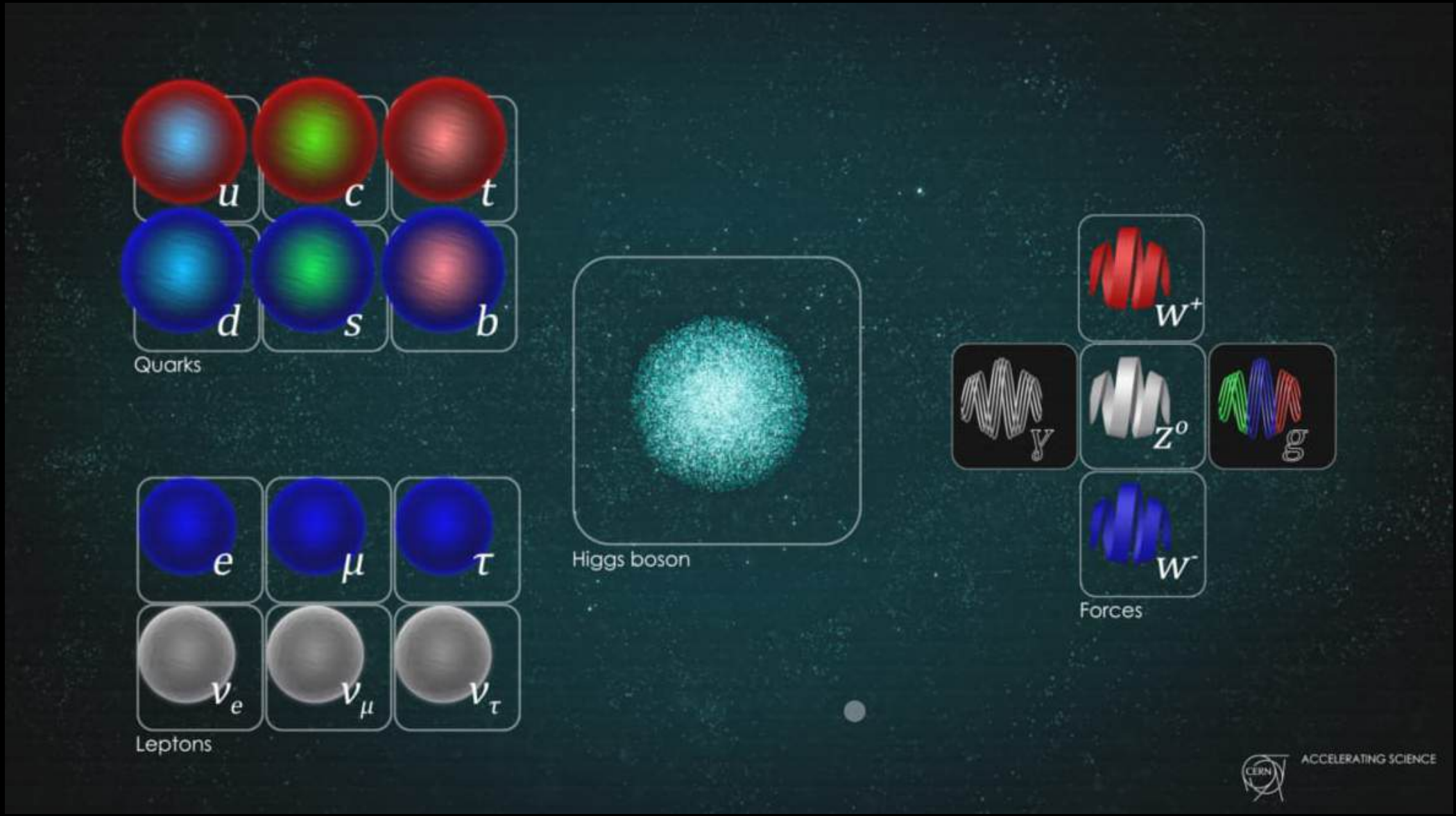
QUANTUM FIELD THEORY

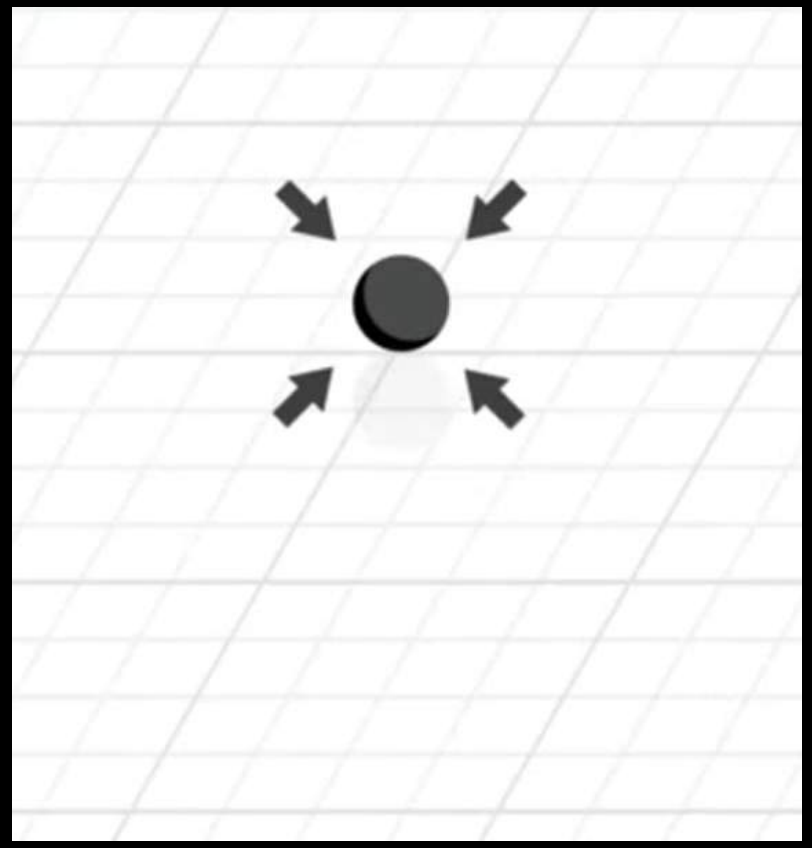
LIMITATIONS OF QM	HOW QFT TACKLES THEM
Fixed particle number	Fields allow creation & annihilation
Non relativistic	Lorentz-invariant field equations
Negative energies	Interpreted as antiparticles
Forces as potentials	Forces as particle exchanges
Vacuum is empty	Vacuum filled with fluctuating fields

EXTENSION OF QUANTUM MECHANICS BY INCLUDING RELATIVITY

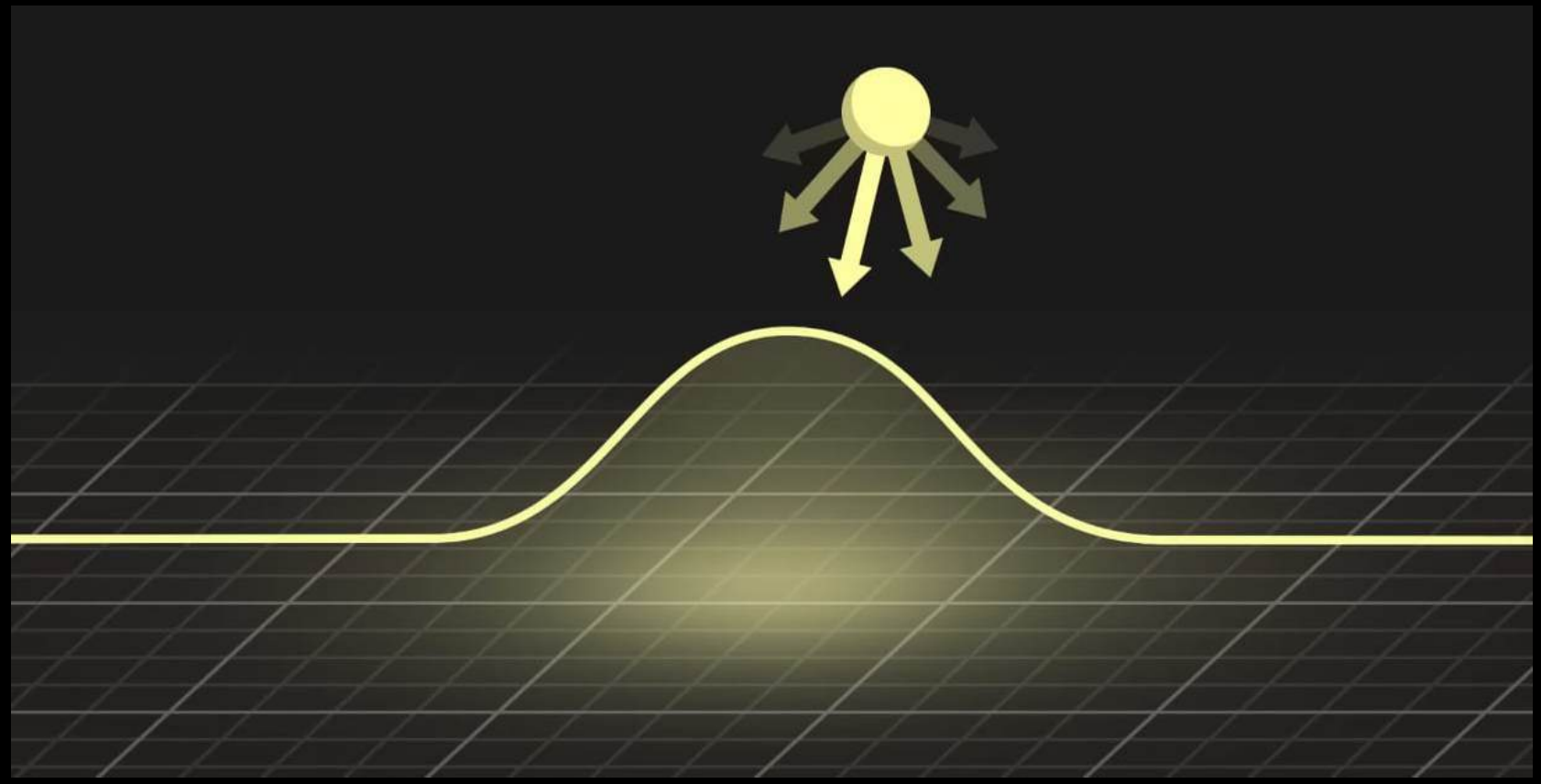


STANDARD MODEL OF PARTICLE PHYSICS

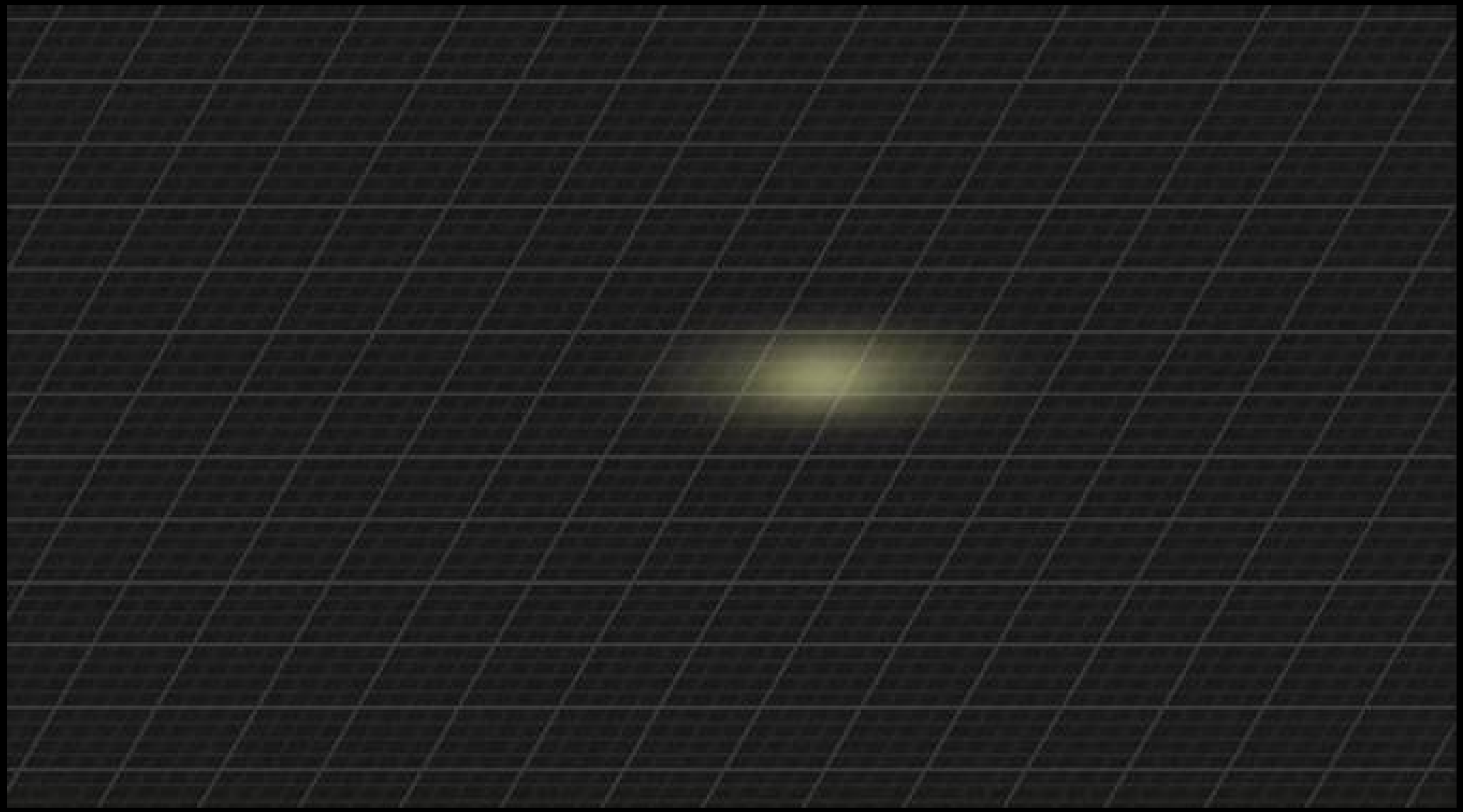


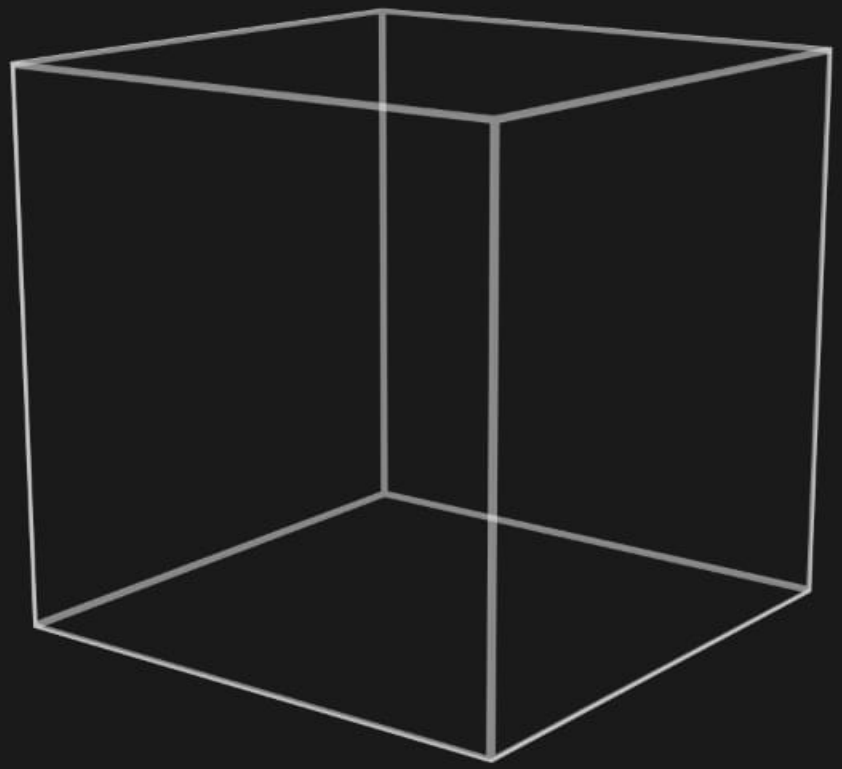


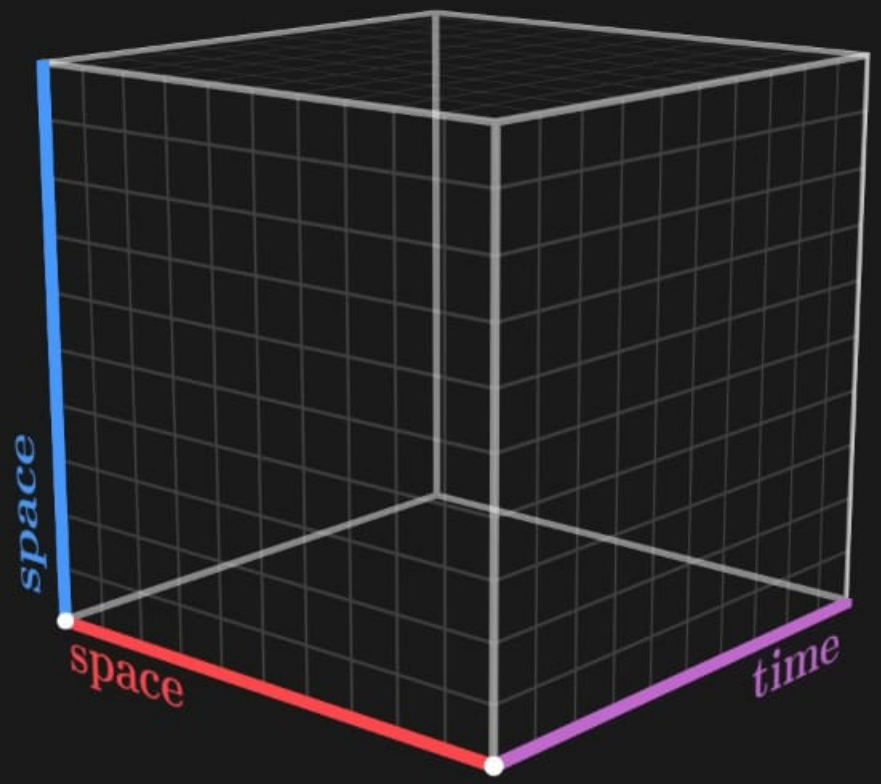
PARTICLE IN CM



PARTICLE IN QM



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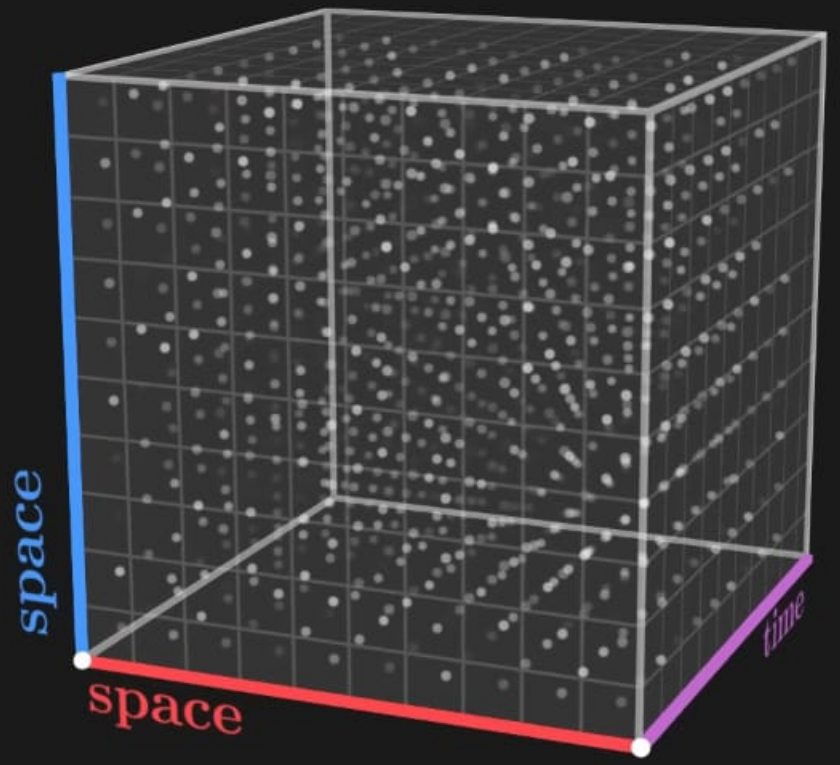
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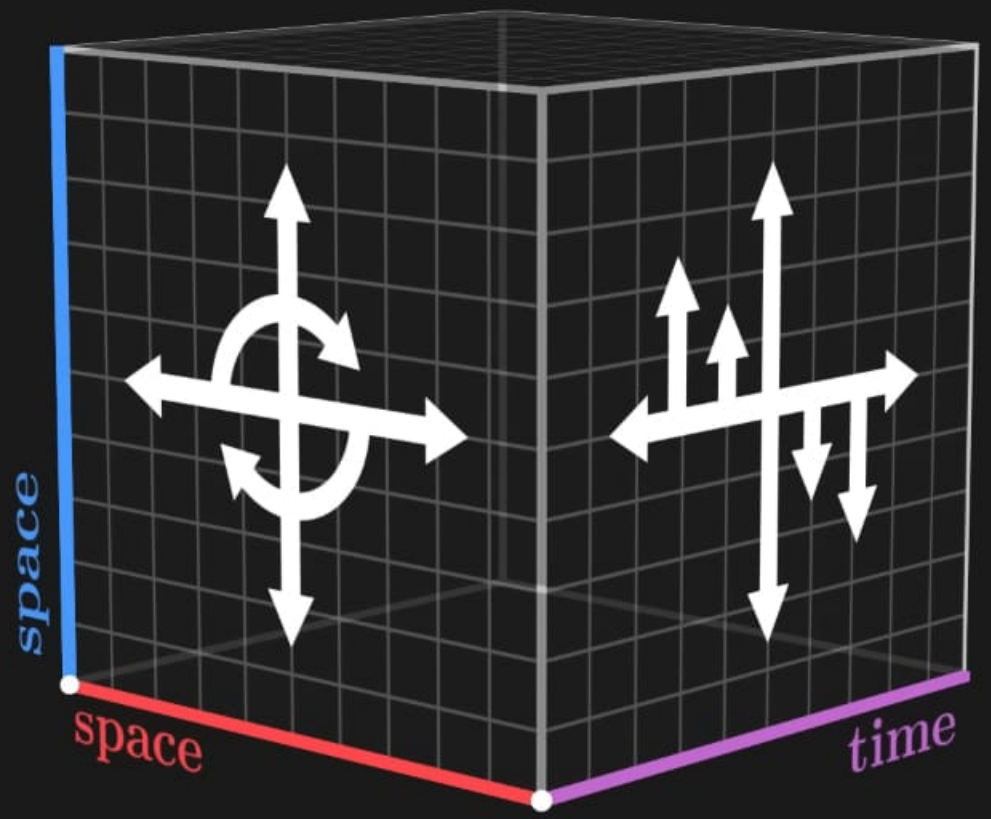
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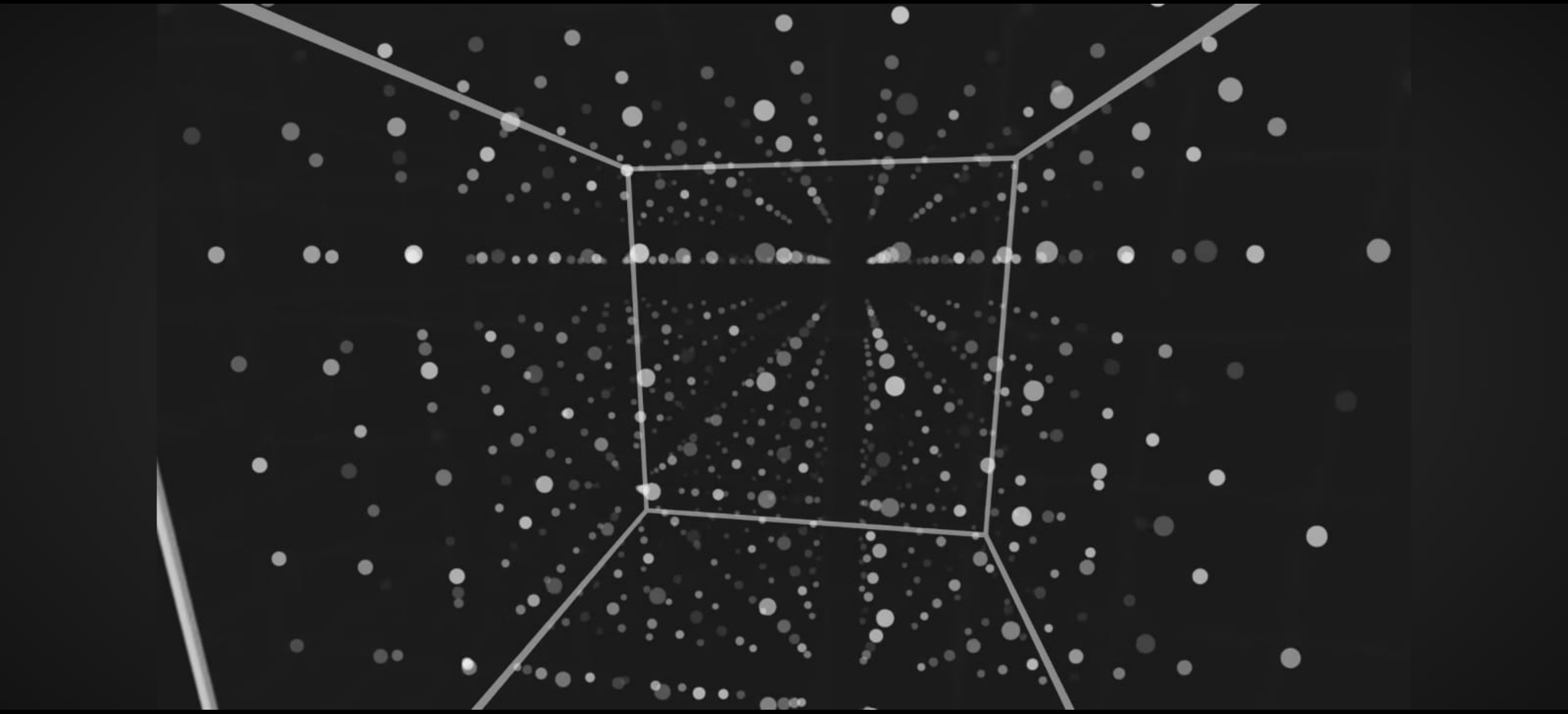
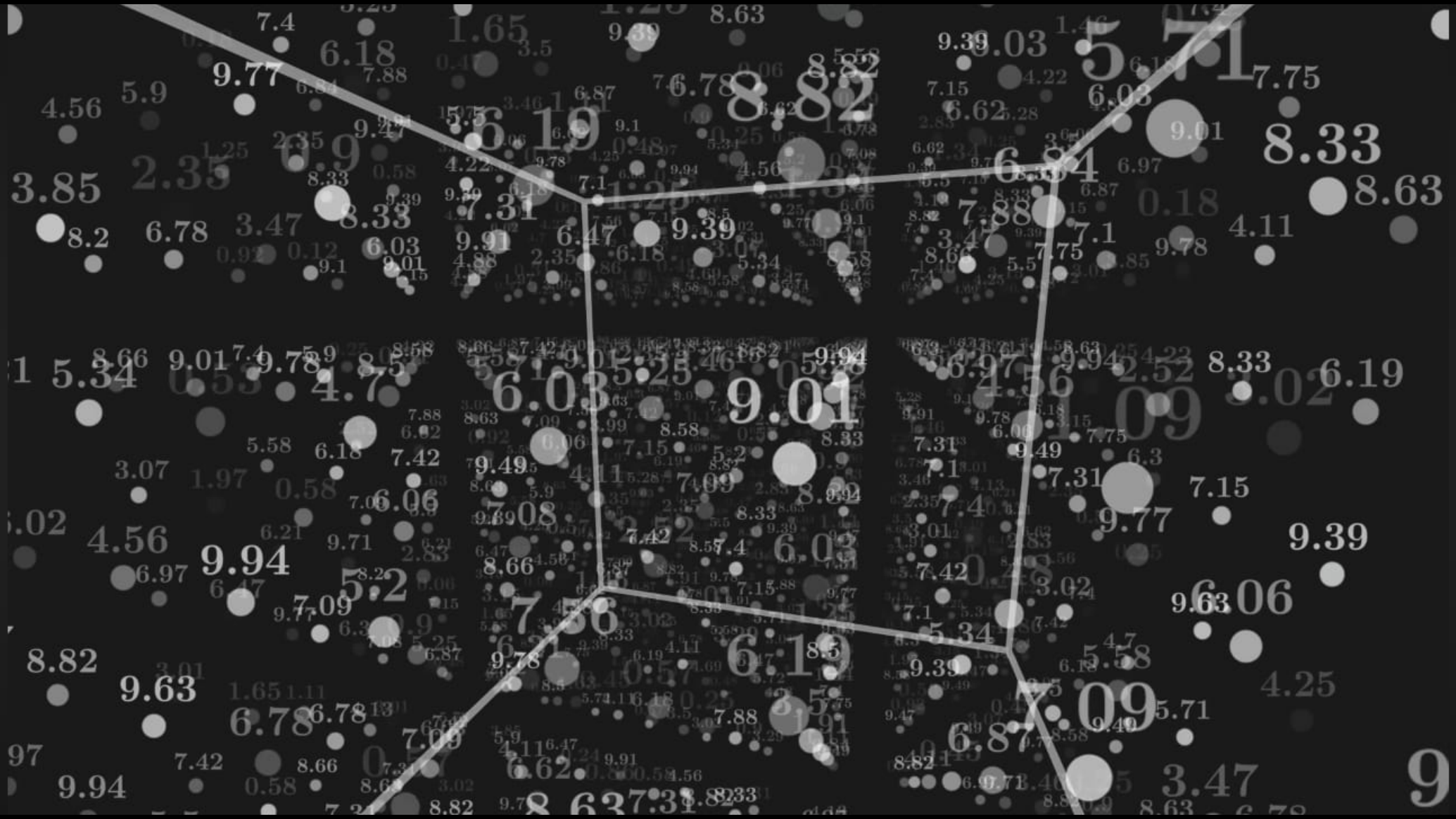
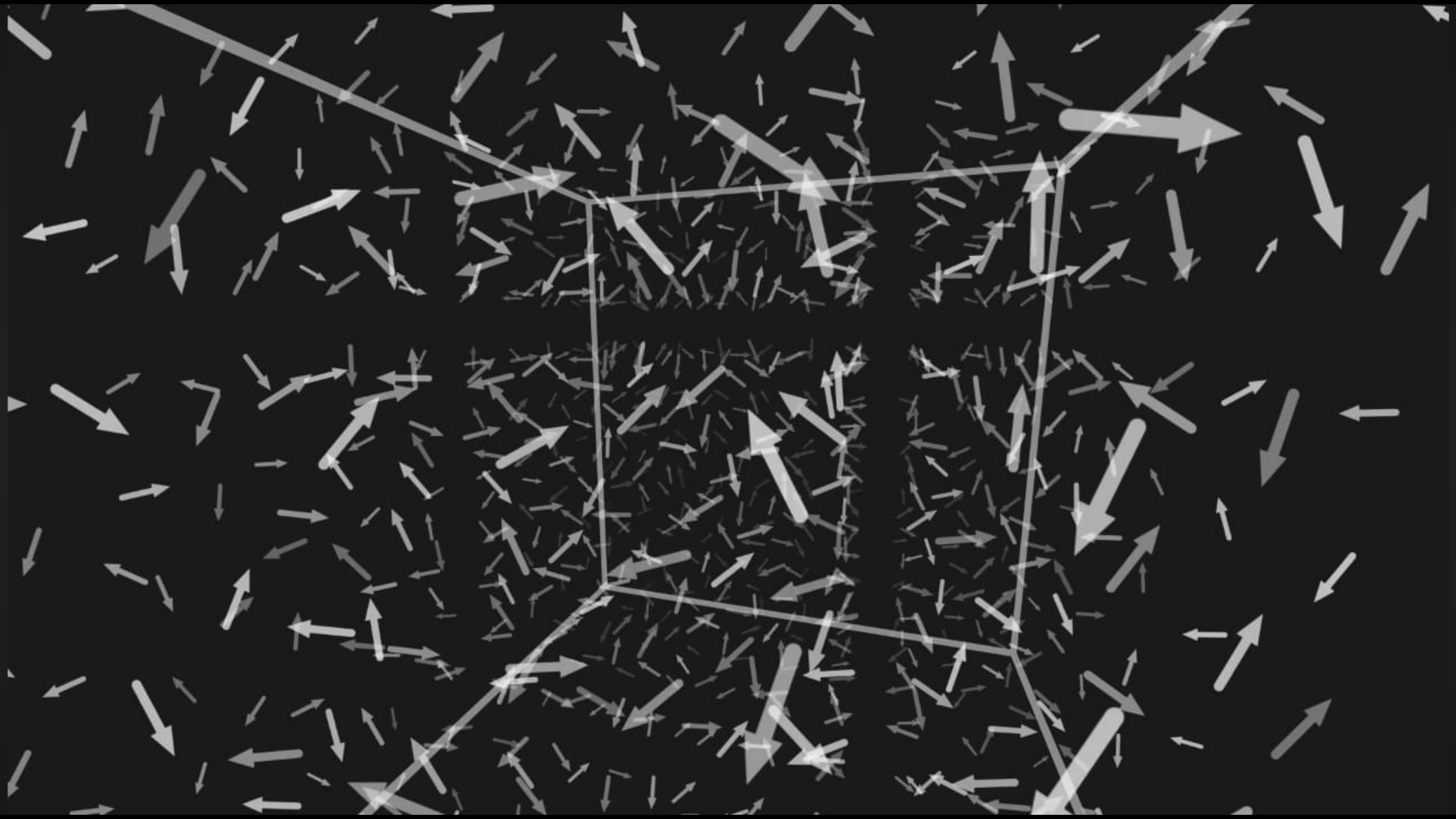
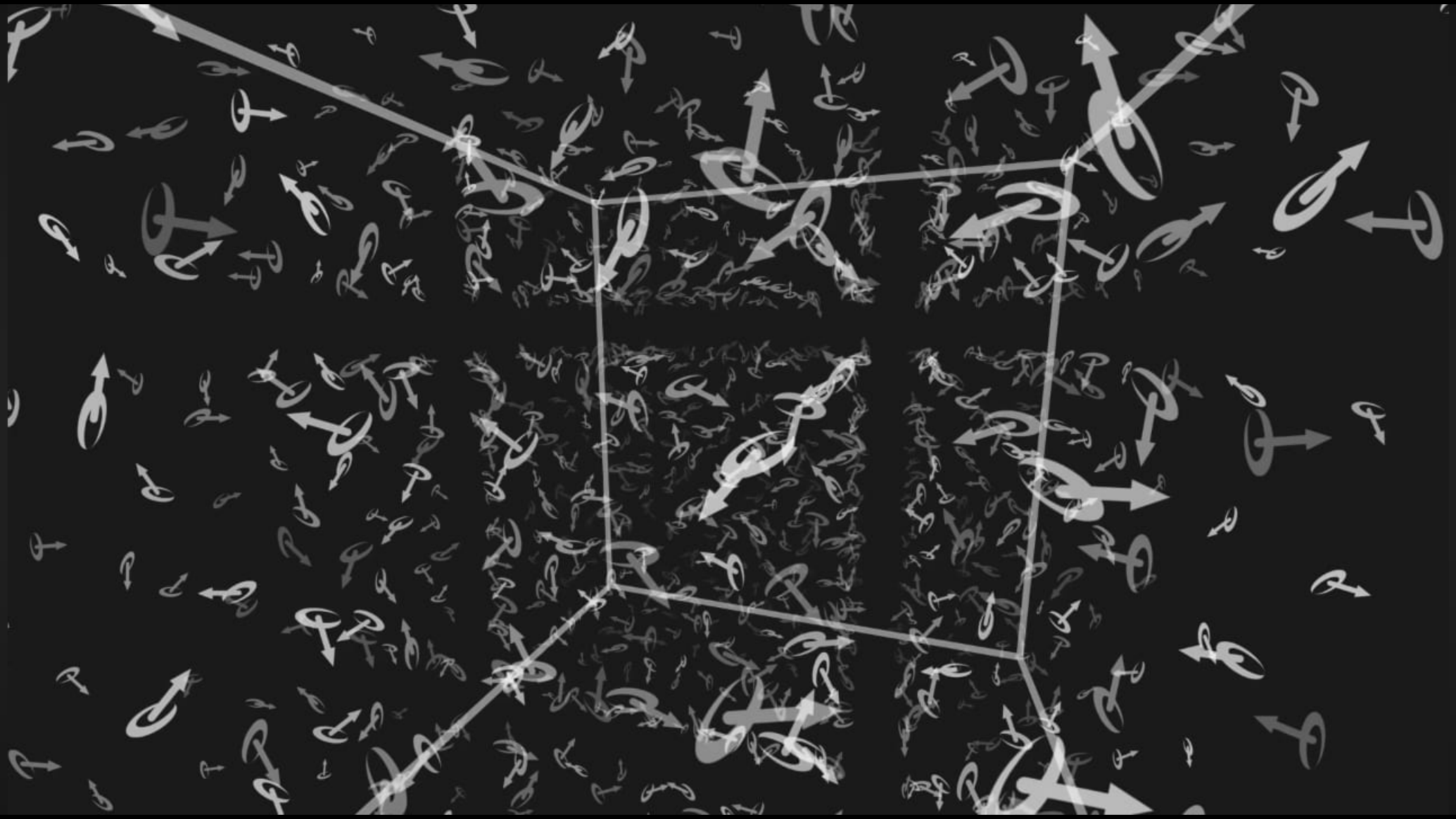
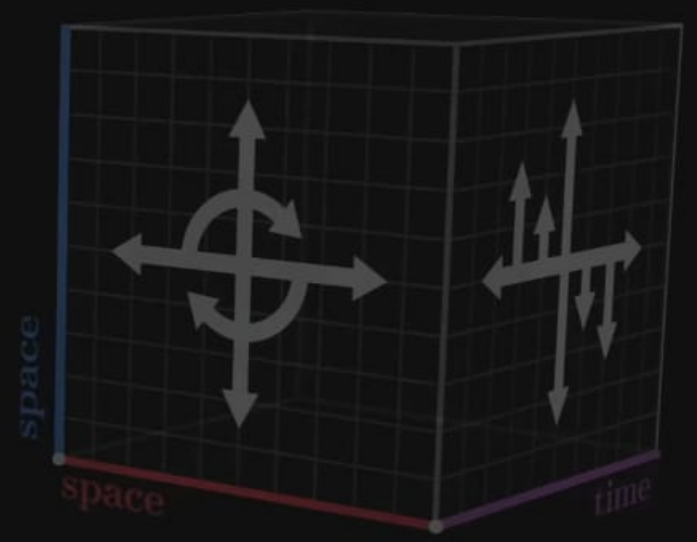
A CLASSICAL FIELD ASSIGNS A SPECIFIC VALUE (EG, A NUMBER, A VECTOR OR ANY OTHER MATHEMATICAL OBJECT TO EVERY POINT IN SPACE & TIME







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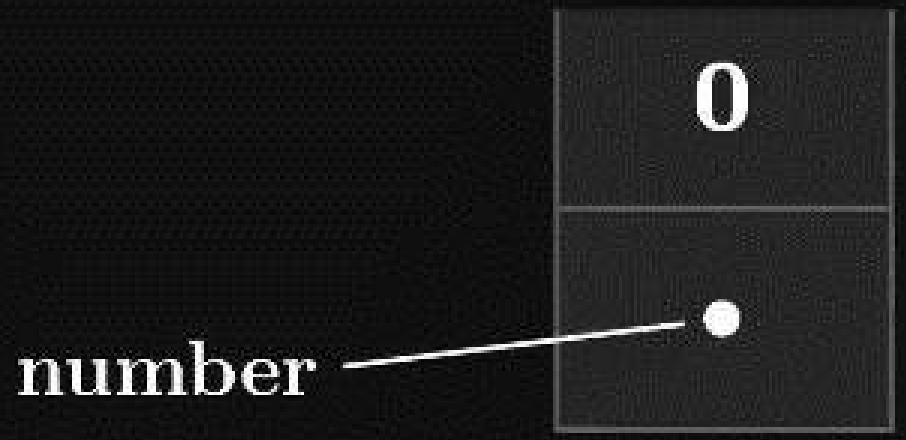
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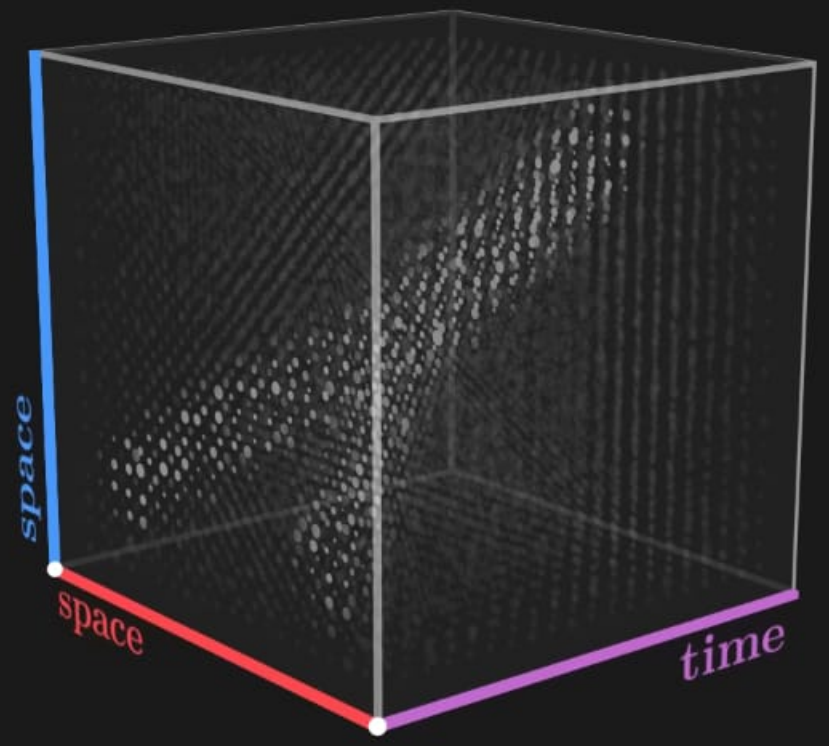
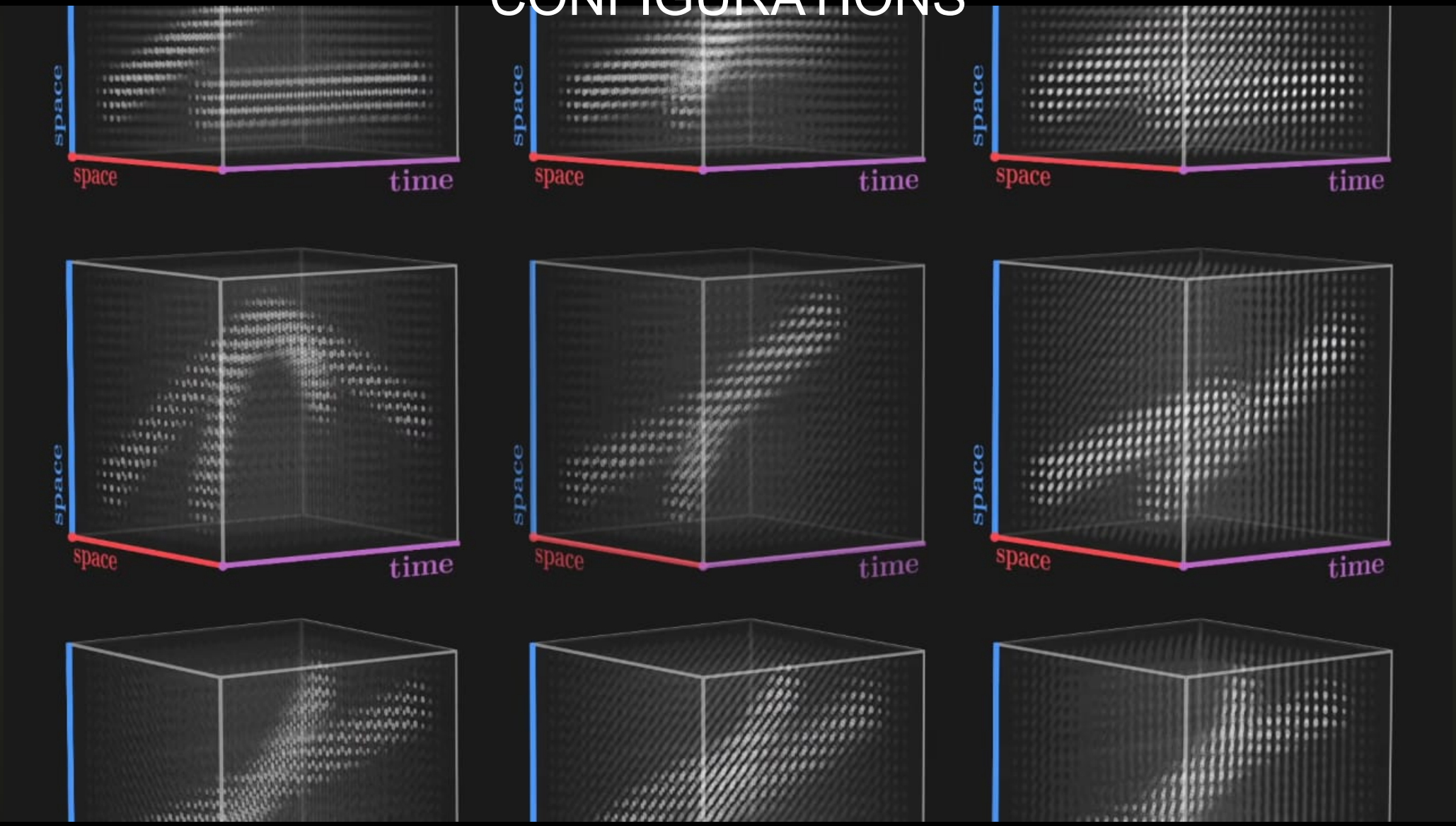
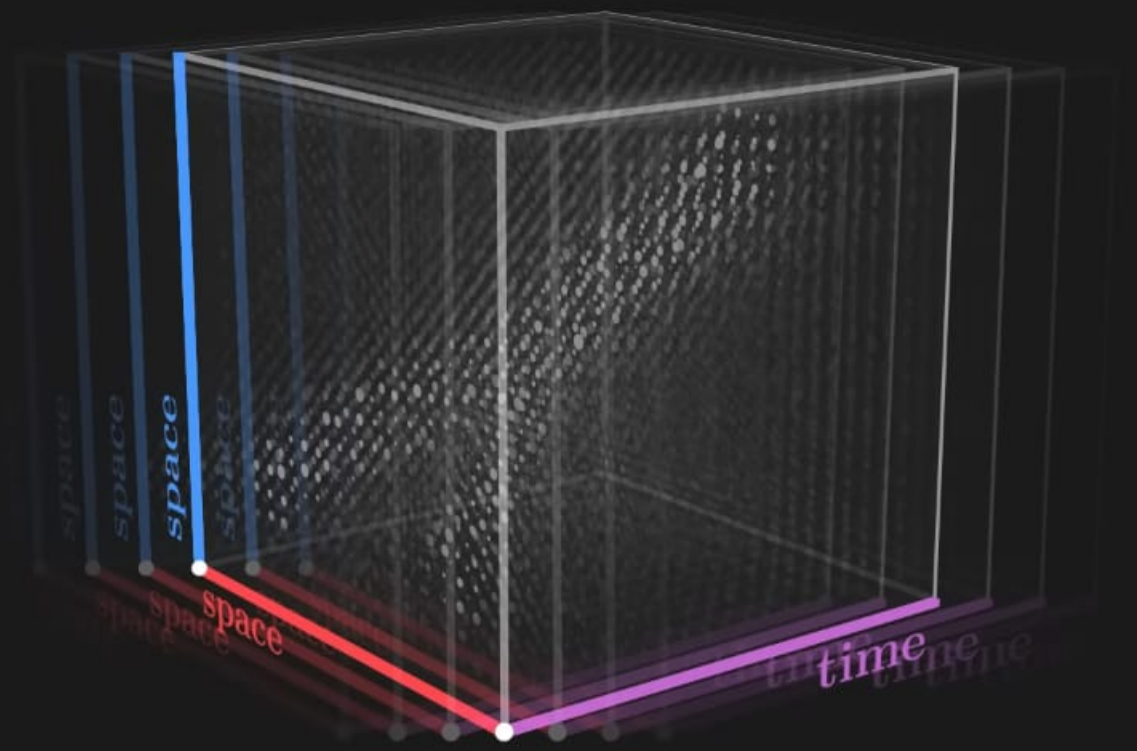
CLASSICAL FIELD HAS A SINGLE CONFIGURATION



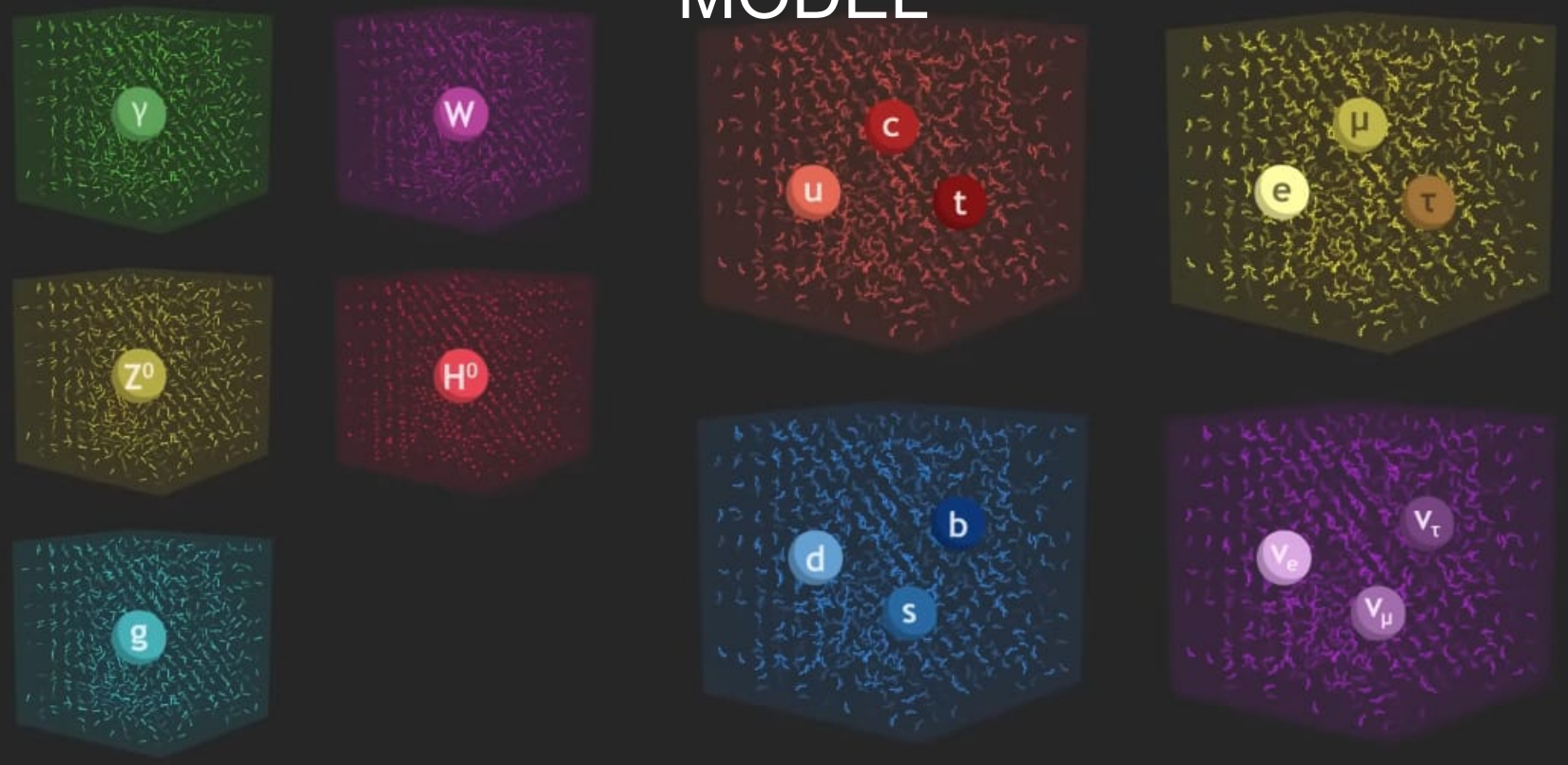
QUANTUM FIELD REPRESENTS A STATE OF ALL POSSIBLE CONFIGURATIONS

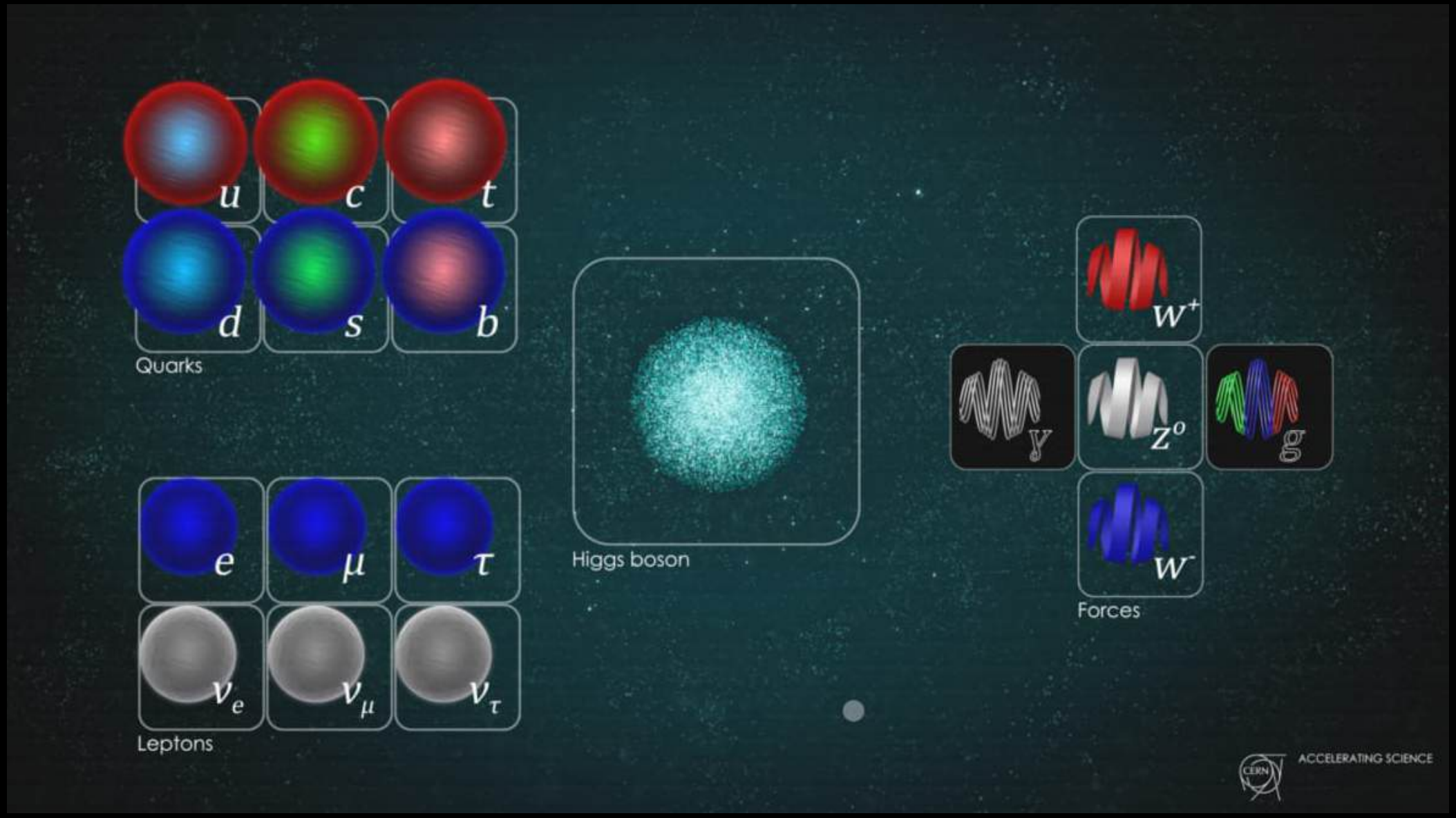


A QUANTUM FIELD ASSIGNS A QUANTUM OPERATOR TO EVERY POINT IN SPACE & TIME



TYPES OF QUANTUM FIELDS CORRESPONDING TO EACH TYPE OF FUNDAMENTAL PARTICLE IN THE STANDARD MODEL

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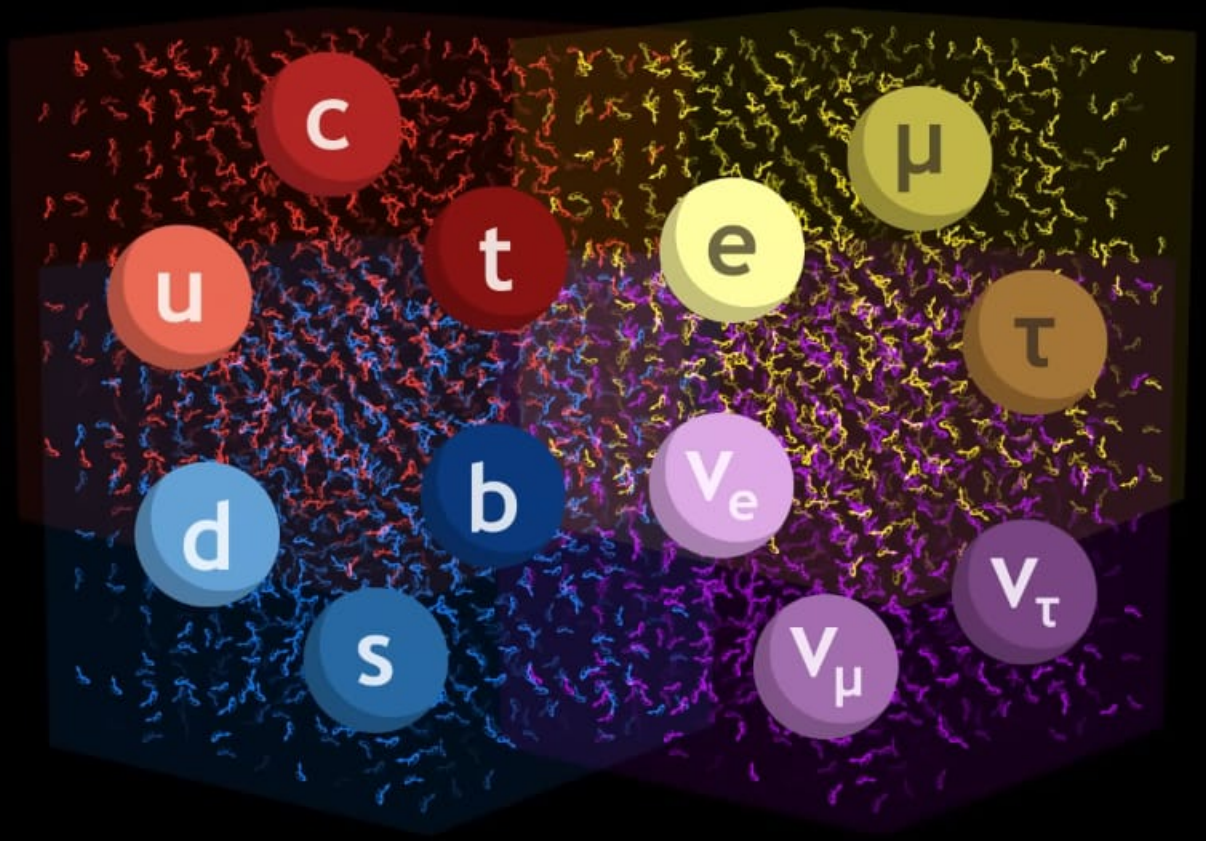
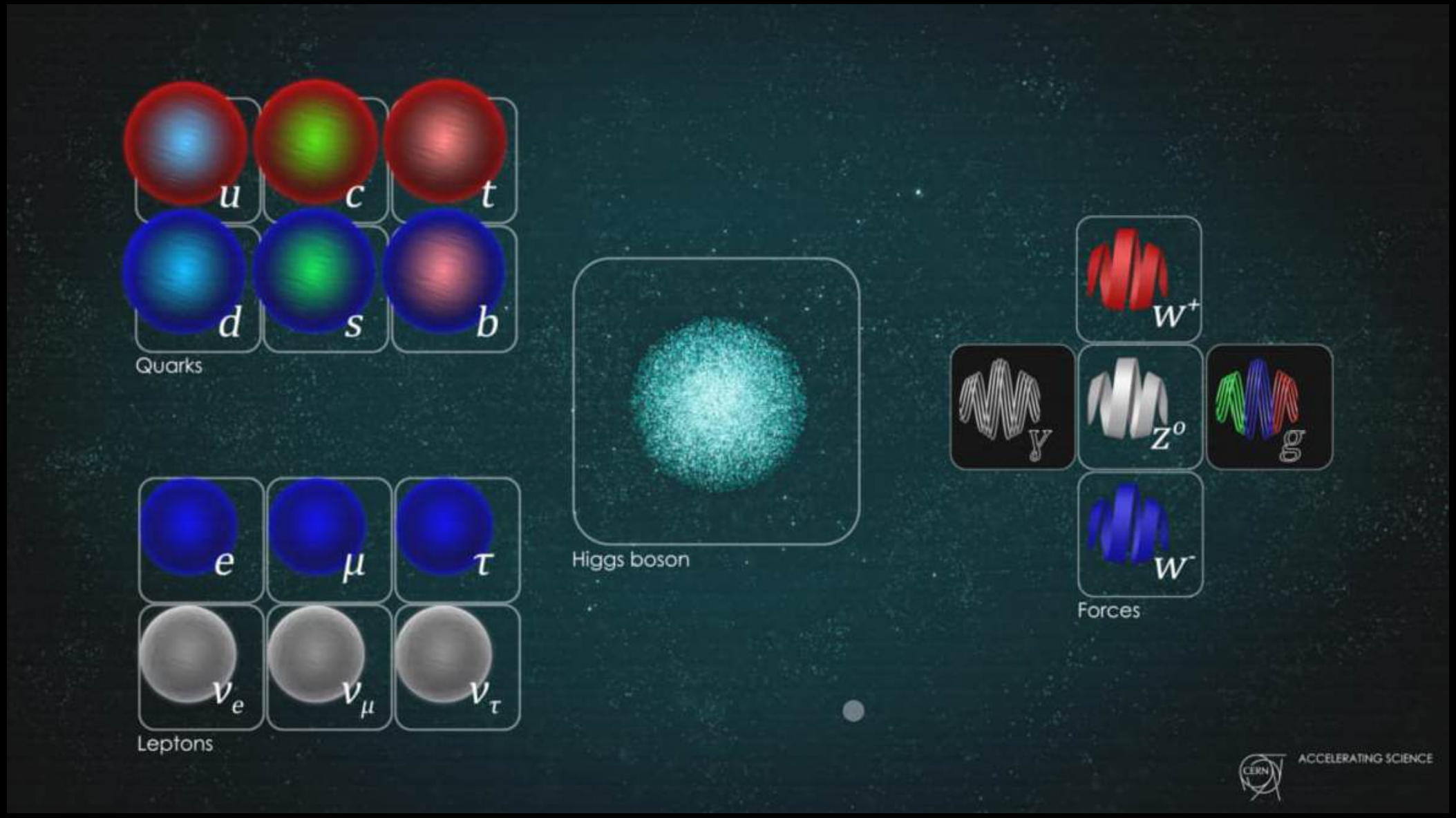
ACCELERATING SCIENCE

FERMION FIELD

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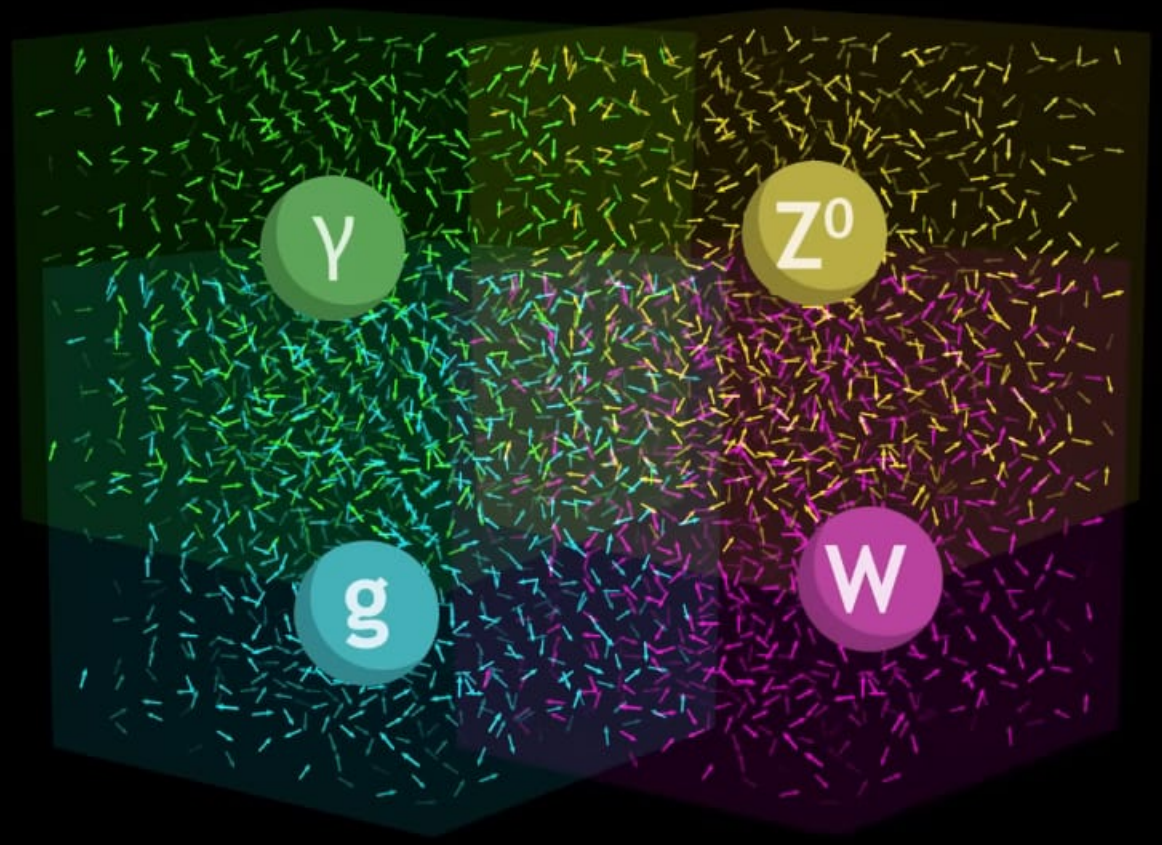
VECTOR FIELD



SCALAR FIELD

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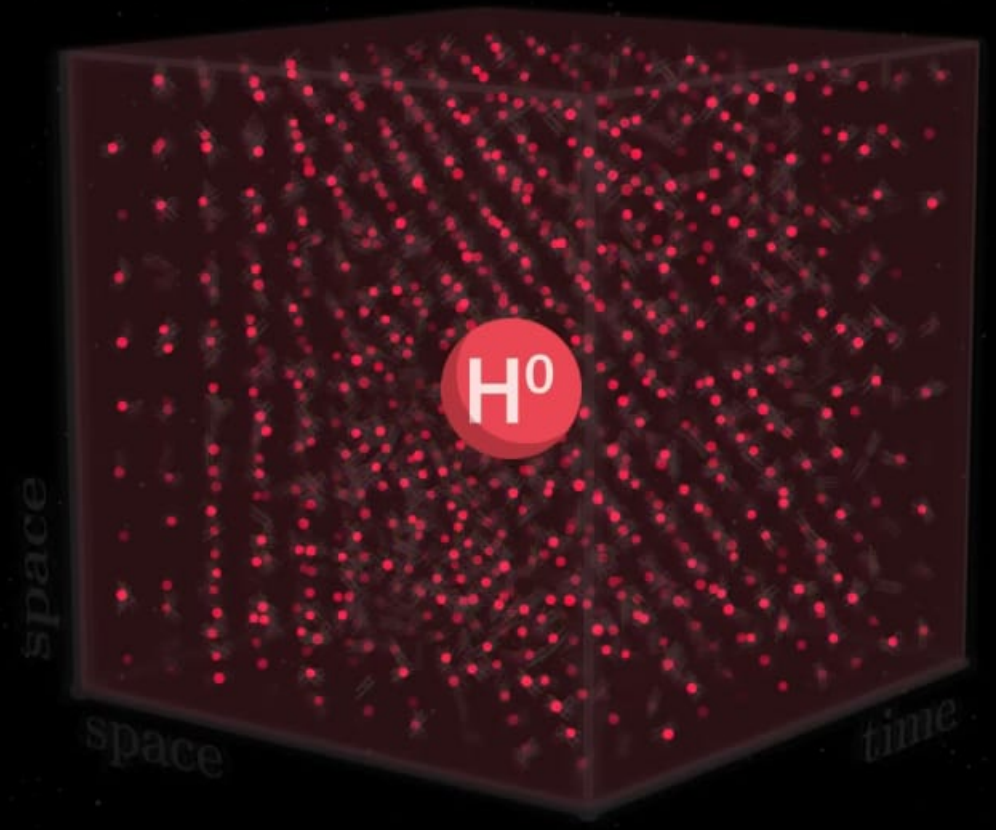
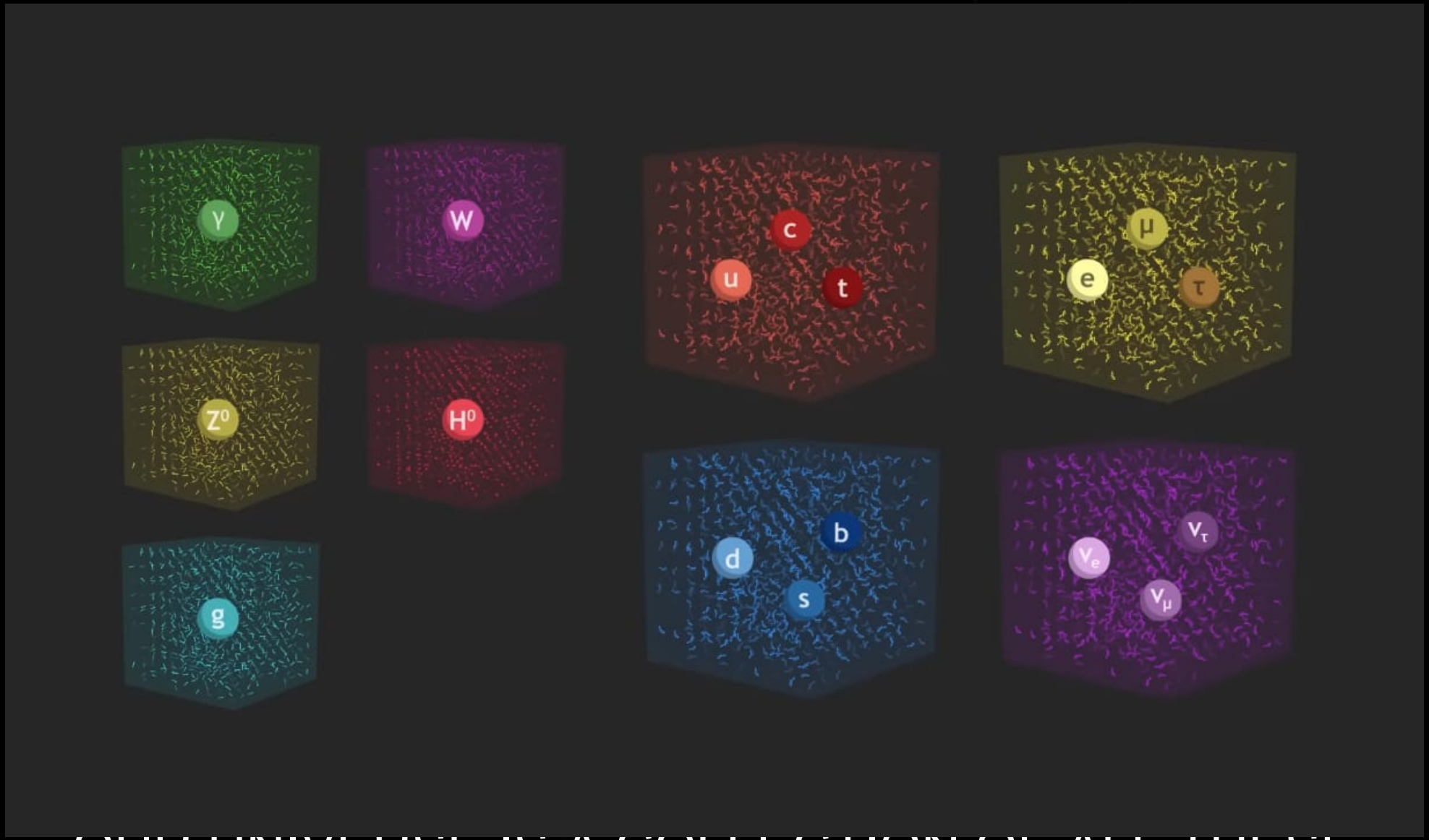
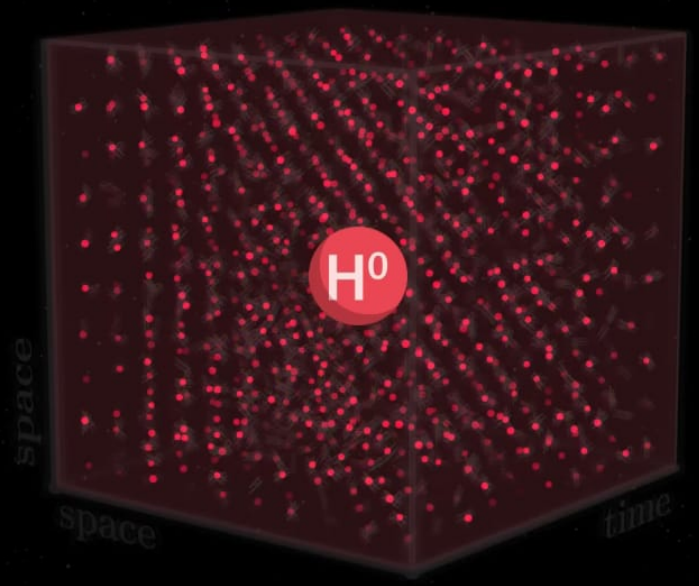




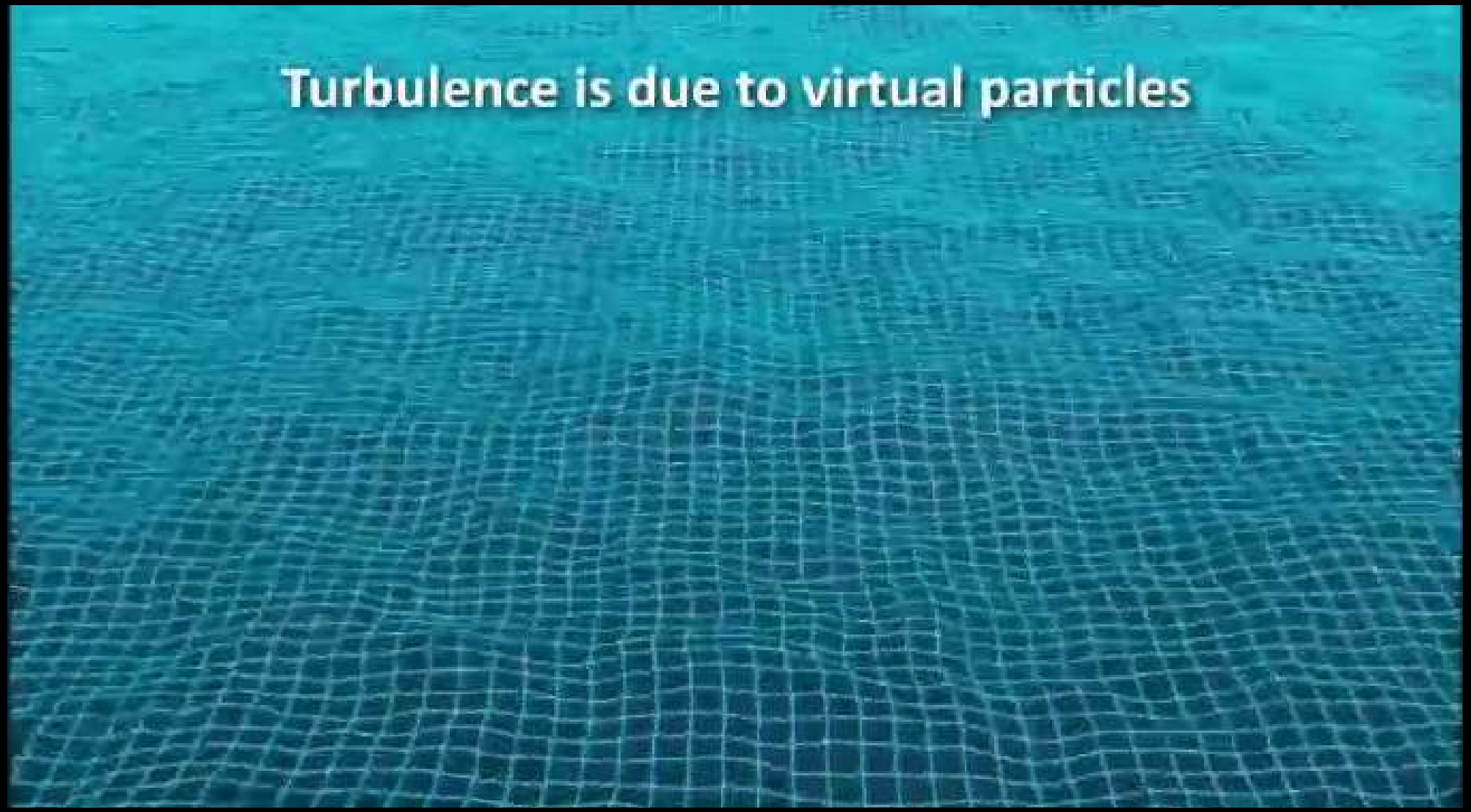
OUR UNIVERSE IS A COLLECTION OF ALL THESE QUANTUM FIELDS

PARTICLES ACQUIRE MASS BY INTERACTING WITH HIGGS FIELD



EXPLAINS WHY PHOTONS WHICH DO NOT INTERACT WITH HIGGS FIELD ARE MASSLESS

PARTICLES ARE EXCITATIONS IN A FIELD



HOW FIELDS ACCOUNT FOR CREATION & ANNIHILATION OF FUNDAMENTAL PARTCLES:

SOURCE
https://youtu.be/MmG2ah5Df4g?si=aBvD0RilEvW0E2U1
https://youtu.be/eoStndCzFhg?si=8LjWMijyt4JIU3tP